



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA	Div: B	Course Code: MCA01604	Batch: S2
Semester: III		Course Name: Data Science Laboratory	
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CO No: CO605.2		Assignment No: 07	

Title: Using grocery dataset with minimum support to 0.001 and minimum confidence of 0.8 build a Frequent Pattern Tree (FP-Tree). Show for each transaction how the tree evolves

Code:

```
# --- Step 1: Import necessary libraries ---
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import fpgrowth, association_rules
from tabulate import tabulate

# --- Step 2: Data Loading and Preparation ---
filename = 'groceries.csv'
dataset = []

try:
    df = pd.read_csv(filename)

    df = df.drop(columns=df.columns[0])

    for index, row in df.iterrows():
        transaction = row.dropna().tolist()
        dataset.append(transaction)

except FileNotFoundError:
    print(f"Error: '{filename}' not found. Please make sure the file is in the same directory as the script.")
    exit()

except Exception as e:
    print(f"An error occurred: {e}")
    exit()

te = TransactionEncoder()
te_ary = te.fit(dataset).transform(dataset)
df_encoded = pd.DataFrame(te_ary, columns=te.columns_)

# --- Step 3: Apply the FP-Growth algorithm with your specified support ---
min_support_threshold = 0.01
```



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```
frequent_itemsets = fpgrowth(df_encoded, min_support=min_support_threshold, use_colnames=True)

# --- Step 4: Generate association rules with your specified confidence ---
min_confidence_threshold = 0.5
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=min_confidence_threshold)

# --- Step 5: Display Output ---
print("\n" + "="*80)
print(" " * 20 + "FP-GROWTH ANALYSIS RESULTS")
print("="*80 + "\n")

# --- Frequent Itemsets ---
print(f"Top 15 Frequent Itemsets (Support >= {min_support_threshold})")
sorted_itemsets = frequent_itemsets.sort_values(by='support', ascending=False)
print(tabulate(sorted_itemsets.head(15), headers='keys', tablefmt='psql', showindex=False))
print(f"\n[INFO] Found a total of {len(frequent_itemsets)} frequent itemsets.\n")

# --- Association Rules ---
print(f"Generated Association Rules (Confidence >= {min_confidence_threshold})")
if rules.empty:
    print("No rules found for the given support and confidence thresholds.")
else:
    relevant_rules = rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']]
    sorted_rules = relevant_rules.sort_values(by=['confidence', 'lift'], ascending=False)
    print(tabulate(sorted_rules.head(15), headers='keys', tablefmt='psql', showindex=False, floatfmt=".4f"))
    print(f"\n[INFO] Found a total of {len(rules)} rules.\n")
```



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Output:

```
=====
FP-GROWTH ANALYSIS RESULTS
=====

Top 15 Frequent Itemsets (Support >= 0.01)
-----+-----+-----
| support | itemsets |
-----+-----+-----
| 0.255516 | frozenset({'whole milk'}) |
| 0.193493 | frozenset({'other vegetables'}) |
| 0.183935 | frozenset({'rolls/buns'}) |
| 0.174377 | frozenset({'soda'}) |
| 0.139502 | frozenset({'yogurt'}) |
| 0.110524 | frozenset({'bottled water'}) |
| 0.108998 | frozenset({'root vegetables'}) |
| 0.104931 | frozenset({'tropical fruit'}) |
| 0.0985257 | frozenset({'shopping bags'}) |
| 0.0939502 | frozenset({'sausage'}) |
| 0.088968 | frozenset({'pastry'}) |
| 0.0827656 | frozenset({'citrus fruit'}) |
| 0.0805287 | frozenset({'bottled beer'}) |
| 0.079817 | frozenset({'newspapers'}) |
| 0.0776817 | frozenset({'canned beer'}) |
-----+-----+-----

[INFO] Found a total of 333 frequent itemsets.

Generated Association Rules (Confidence >= 0.5)
-----+-----+-----+-----+-----+-----+
| antecedents | consequents | support | confidence | lift |
-----+-----+-----+-----+-----+-----+
| frozenset({'root vegetables', 'citrus fruit'}) | frozenset({'other vegetables'}) | 0.0104 | 0.5862 | 3.0296 |
| frozenset({'root vegetables', 'tropical fruit'}) | frozenset({'other vegetables'}) | 0.0123 | 0.5845 | 3.0210 |
| frozenset({'yogurt', 'curd'}) | frozenset({'whole milk'}) | 0.0101 | 0.5824 | 2.2791 |
| frozenset({'other vegetables', 'butter'}) | frozenset({'whole milk'}) | 0.0115 | 0.5736 | 2.2449 |
| frozenset({'root vegetables', 'tropical fruit'}) | frozenset({'whole milk'}) | 0.0120 | 0.5700 | 2.2310 |
| frozenset({'yogurt', 'root vegetables'}) | frozenset({'whole milk'}) | 0.0145 | 0.5630 | 2.2034 |
| frozenset({'other vegetables', 'domestic eggs'}) | frozenset({'whole milk'}) | 0.0123 | 0.5525 | 2.1623 |
| frozenset({'yogurt', 'whipped/sour cream'}) | frozenset({'whole milk'}) | 0.0109 | 0.5245 | 2.0527 |
| frozenset({'root vegetables', 'rolls/buns'}) | frozenset({'whole milk'}) | 0.0127 | 0.5230 | 2.0469 |
| frozenset({'other vegetables', 'pip fruit'}) | frozenset({'whole milk'}) | 0.0135 | 0.5175 | 2.0254 |
| frozenset({'yogurt', 'tropical fruit'}) | frozenset({'whole milk'}) | 0.0151 | 0.5174 | 2.0248 |
| frozenset({'other vegetables', 'yogurt'}) | frozenset({'whole milk'}) | 0.0223 | 0.5129 | 2.0072 |
| frozenset({'other vegetables', 'whipped/sour cream'}) | frozenset({'whole milk'}) | 0.0146 | 0.5070 | 1.9844 |
| frozenset({'root vegetables', 'rolls/buns'}) | frozenset({'other vegetables'}) | 0.0122 | 0.5021 | 2.5949 |
| frozenset({'yogurt', 'root vegetables'}) | frozenset({'other vegetables'}) | 0.0129 | 0.5000 | 2.5841 |
-----+-----+-----+-----+-----+-----+

[INFO] Found a total of 15 rules.
```

Each transaction how the tree evolves

1. Calculate Item Frequency & Establish Order

- bread: 5
- milk: 4
- butter: 4
- sugar: 2
- eggs: 2
- coffee: 1 (Discarded as its count is less than 2)



2. Building the FP-Tree Transaction by Transaction

Transaction 1:

```
Tree State:
(root)
└─ bread:1
   └─ butter:1
      └─ milk:1
```

Transaction 2:

```
Tree State:
(root)
└─ bread:2
   ├── butter:1
   │   └─ milk:1
   └─ milk:1
       └─ sugar:1
```

Transaction 3:

```
Tree State:
(root)
├─ bread:2
│   ├── butter:1
│   │   └─ milk:1
│   └─ milk:1
│       └─ sugar:1
└─ butter:1
    └─ milk:1
        └─ eggs:1
```

Transaction 4:

```
Tree State:
(root)
├─ bread:3
│   ├── butter:2
│   │   ├── milk:1
│   │   └─ eggs:1
│   └─ milk:1
│       └─ sugar:1
└─ butter:1
    └─ milk:1
        └─ eggs:1
```



Transaction 5:

```
Tree State:
(root)
├─ bread:4
│   ├── butter:3
│   │   ├── milk:2
│   │   │   └─ sugar:1
│   │   └─ eggs:1
│   └─ milk:1
│       └─ sugar:1
└─ butter:1
    └─ milk:1
        └─ eggs:1
```

3. FP-Tree Structure

```
Final Tree:
(root)
├─ bread:5
│   ├── butter:3
│   │   ├── milk:2
│   │   │   └─ sugar:1
│   │   └─ eggs:1
│   └─ milk:1
│       └─ sugar:1
└─ butter:1
    └─ milk:1
        └─ eggs:1
```